H1N1-VACCINE-ANALYSIS-PREDICTIONS

Stakeholder

Public Health Authorities

Problem Statement

Public health authorities need to increase vaccination uptake to reduce the spread of infectious diseases and protect vulnerable population. However, many individuals choose not to receive vaccines due to differences in beliefs, health behaviors, risk perception and access to healthcare. Without understanding which groups are less likely to vaccinate, it is diicult for health organisation to design effective outreach and vaccination campaigns

Business Understanding

KEY QUESTION FOR PREDICTING H1N1 VACCINE UPTAKE

1.Which factors influence whether an individual receives the H1N1 flu vaccine and can we accurately predict who is likely or unilikely to get vaccinated?

2 Which demographic group are les likely to vaccinate? Understanding differences in vaccination behavior across age, gender, education level and income helps identify high risk populations.

3.Does doctor recommendation influences vaccination behavior? Medical professionals are trusted sources of health advice. Analysing whether individuals who receives recommendations from their doctors are more likely to vaccinate can guide strategies for improving outreach through healthcare providers.

4,Which health condition are associated with vaccination? People with chronic medical condition may be more motivated to vaccinate. identifying these patterns helps príoritize interventions for high risk groups.

Import Libraries

Importing libraries for data manupulation,visualization,machine learning, preprocessing and evalution.

In [3]:
```
# Data manipulation
import pandas as pd
import numpy as np

# Visualization
```

```
import matplotlib.pyplot as plt
import seaborn as sns

# Machine learning models
from sklearn.model_selection import train_test_split, cross_val_score, GridSearc
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier

# Preprocessing
from sklearn.preprocessing import OneHotEncoder
from sklearn.impute import SimpleImputer
from sklearn.pipeline import Pipeline
from sklearn.compose import ColumnTransformer

# Evaluation
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_sc
```

Load the Dataset

Load the dataset into a DataFrame and show the first 5 rows so that we can inspect the data

In [4]:
```
df = pd.read_csv("H1N1_Flu_vaccines.csv")
df.head()
```

Out[4]:

	respondent_id	h1n1_concern	h1n1_knowledge	behavioral_antiviral_meds	behavioral
0	0	1.0	0.0	0.0	
1	1	3.0	2.0	0.0	
2	2	1.0	1.0	0.0	
3	3	1.0	1.0	0.0	
4	4	2.0	1.0	0.0	

5 rows × 38 columns

Check Columns

This helps us see all the variables in the dataset and identfy the target columns, feature columns and unnecessary columns

In [5]:
```
print(df.columns.tolist())
```

```
['respondent_id', 'h1n1_concern', 'h1n1_knowledge', 'behavioral_antiviral_meds',
'behavioral_avoidance', 'behavioral_face_mask', 'behavioral_wash_hands', 'behavio
ral_large_gatherings', 'behavioral_outside_home', 'behavioral_touch_face', 'docto
r_recc_h1n1', 'doctor_recc_seasonal', 'chronic_med_condition', 'child_under_6_mon
ths', 'health_worker', 'health_insurance', 'opinion_h1n1_vacc_effective', 'opinio
n_h1n1_risk', 'opinion_h1n1_sick_from_vacc', 'opinion_seas_vacc_effective', 'opin
ion_seas_risk', 'opinion_seas_sick_from_vacc', 'age_group', 'education', 'race',
'sex', 'income_poverty', 'marital_status', 'rent_or_own', 'employment_status', 'h
hs_geo_region', 'census_msa', 'household_adults', 'household_children', 'employme
nt_industry', 'employment_occupation', 'h1n1_vaccine', 'seasonal_vaccine']
```

Check Dataset Shape and Missing Vlues

This is important because missing values must be handled before modeling

In [6]:
```
print(df.shape)
print(df.isnull().sum())
```

```
(26707, 38)
respondent_id                     0
h1n1_concern                     92
h1n1_knowledge                  116
behavioral_antiviral_meds        71
behavioral_avoidance            208
behavioral_face_mask             19
behavioral_wash_hands            42
behavioral_large_gatherings      87
behavioral_outside_home          82
behavioral_touch_face           128
doctor_recc_h1n1               2160
doctor_recc_seasonal           2160
chronic_med_condition           971
child_under_6_months            820
health_worker                   804
health_insurance              12274
opinion_h1n1_vacc_effective     391
opinion_h1n1_risk               388
opinion_h1n1_sick_from_vacc     395
opinion_seas_vacc_effective     462
opinion_seas_risk               514
opinion_seas_sick_from_vacc     537
age_group                         0
education                      1407
race                              0
sex                               0
income_poverty                 4423
marital_status                 1408
rent_or_own                    2042
employment_status              1463
hhs_geo_region                    0
census_msa                        0
household_adults                249
household_children              249
employment_industry           13330
employment_occupation         13470
h1n1_vaccine                      0
seasonal_vaccine                  0
dtype: int64
```

Choose the Target Variable

This project focuses on predicting whether a person received the H1N1 Vaccine. Target Variable would be h1n1_vaccine and we will remove seasonal_vaccine because it is another target variable and keeping it may cause data leakage

In [7]:
```
#Choose H1N1 vaccine as target
target = "h1n1_vaccine"

#features and target
X = df.drop(columns=["h1n1_vaccine","seasonal_vaccine"])
y = df[target]

print("Features shape:", X.shape)
print("Target shape:", y.shape)
```

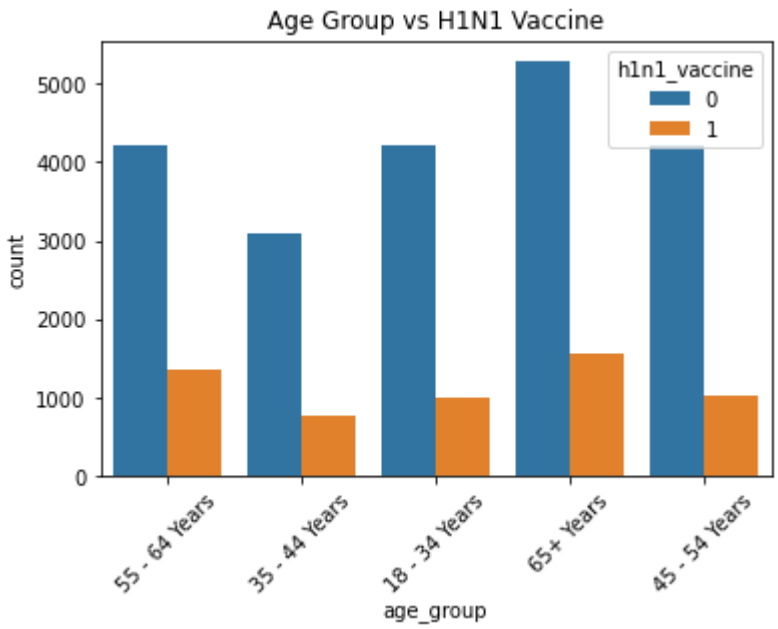
```
Features shape: (26707, 36)
Target shape: (26707,)
```

Data Visualization

In this section we visualize the dataset to better understanding the factors associated with H1N1 vaccination uptake.

Age Group vs H1N1 vaccine

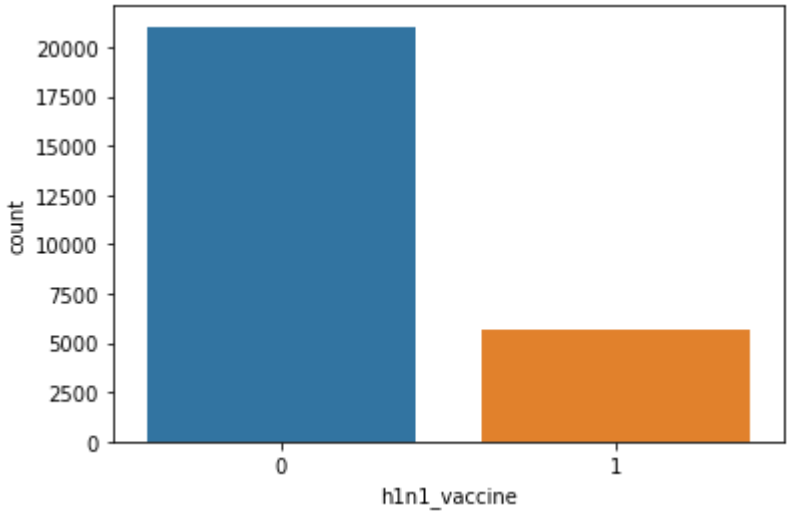
In [8]:
```
plot_df = df[["age_group", "h1n1_vaccine"]].copy()

# Convert to string
plot_df["age_group"] = plot_df["age_group"].astype(str)
plot_df["h1n1_vaccine"] = plot_df["h1n1_vaccine"].astype(str)

# Plot
sns.countplot(x="age_group", hue="h1n1_vaccine", data=plot_df)
plt.xticks(rotation=45)
plt.title("Age Group vs H1N1 Vaccine")
plt.show()
```



Target Distribution

The distribution of the target variable H1N1-Vaccine

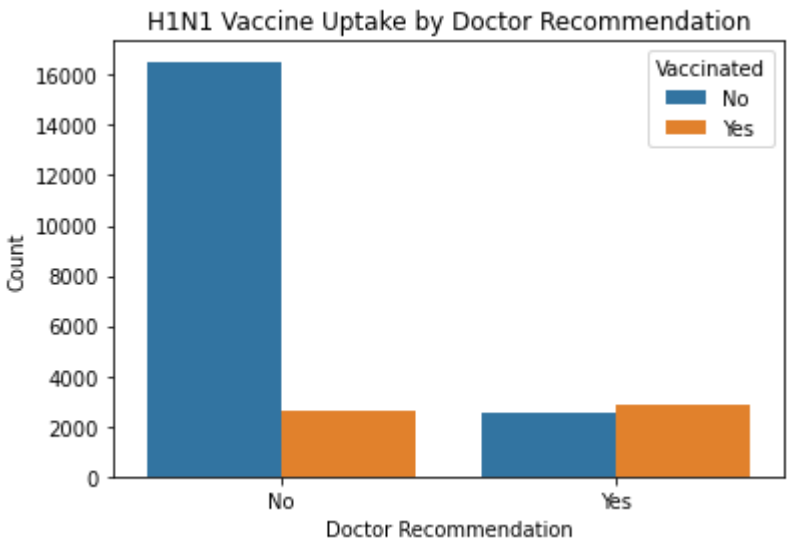
In [9]:
```
sns.countplot(x="h1n1_vaccine", data=df)
plt.show()
```



Doctor reccommendation vs H1N1 vaccine

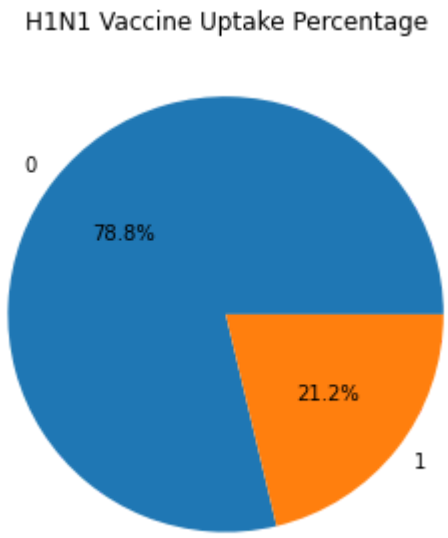
In [10]:
```
# Plot
plot_df = df[["doctor_recc_h1n1", "h1n1_vaccine"]].dropna().copy()

plot_df["doctor_recc_h1n1"] = plot_df["doctor_recc_h1n1"].map({0: "No", 1: "Yes"
plot_df["h1n1_vaccine"] = plot_df["h1n1_vaccine"].map({0: "No", 1: "Yes"})

plt.figure(figsize=(6,4))
sns.countplot(x="doctor_recc_h1n1", hue="h1n1_vaccine", data=plot_df)
plt.title("H1N1 Vaccine Uptake by Doctor Recommendation")
plt.xlabel("Doctor Recommendation")
plt.ylabel("Count")
```

```
plt.legend(title="Vaccinated")
plt.show()
```



H1N1 Vaccine Uptake Percentage

In [11]:
```
# Pie chart
df["h1n1_vaccine"].value_counts().plot(
    kind="pie",
    autopct="%1.1f%%",
    figsize=(5,5)
)
plt.title("H1N1 Vaccine Uptake Percentage")
plt.ylabel("")
plt.show()
```



Train-Test Split

Split the dataset into training data (80%) used to train the model and testing data (20%) used to evaluate the model on unseen data. We used 'stratify=y' to keep same proportion of vaccinated and non-vaccinated individuals in both sets.

In [12]:
```
# Split the data
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42,
    stratify=y
)

print("Training set:", X_train.shape)
print("Testing set:", X_test.shape)
```

```
Training set: (21365, 36)
Testing set: (5342, 36)
```

Identify Numeric and Categorical Columns

We need to preprocess them differently Numeric missing values we fill with median, Cartegorical missing values with most frequent value. Cartegorical variables should be converted into numbers using one-hot encoding.

In [13]:
```
# Separate numeric and categorical columns
numeric_cols = X.select_dtypes(include=["int64", "float64"]).columns.tolist()
categorical_cols = X.select_dtypes(include=["object"]).columns.tolist()

print("Numeric columns:", len(numeric_cols))
print("Categorical columns:", len(categorical_cols))
```

```
Numeric columns: 24
Categorical columns: 12
```

Data Preprocessing

This ensures the machine learning model can work with the data correctly

In [14]:
```
# Preprocessing for numeric data
numeric_transformer = Pipeline(steps=[
    ("imputer", SimpleImputer(strategy="median"))
])

# Preprocessing for categorical data
categorical_transformer = Pipeline(steps=[
    ("imputer", SimpleImputer(strategy="most_frequent")),
    ("onehot", OneHotEncoder(handle_unknown="ignore"))
])

# Combine preprocessing
preprocessor = ColumnTransformer(transformers=[
    ("num", numeric_transformer, numeric_cols),
    ("cat", categorical_transformer, categorical_cols)
])
```

Build Baseline Model -Logistic Regression

We start with logistic regression as the baseline model because it is simple and easy to interpret also it works well for binary classification.

In [15]:
```
# Create baseline model pipeline
log_model = Pipeline(steps=[
    ("preprocessor", preprocessor),
    ("classifier", LogisticRegression(max_iter=1000, random_state=42))
])

# Train the model
log_model.fit(X_train, y_train)

# Predictions
y_train_pred_log = log_model.predict(X_train)
y_test_pred_log = log_model.predict(X_test)
```

Evaluate Logistic Regression

We use the following classification metrics Accuracy to show overall correctness, Precision how many were actually vaccinated, Recall it try to show among actually vaccinated, how many the model correctly identifiend and the last one is F1 Score the balance between precision and recall. We evaluate training data to see how well the model learned and test data to check how well it generalizes to new data.

In [16]:
```
# Function to evaluate model
def evaluate_model(y_true_train, y_pred_train, y_true_test, y_pred_test, model_n
    print(f"\n===== {model_name} =====")

    print("\nTraining Performance")
    print("Accuracy :", round(accuracy_score(y_true_train, y_pred_train), 4))
    print("Precision:", round(precision_score(y_true_train, y_pred_train), 4))
    print("Recall   :", round(recall_score(y_true_train, y_pred_train), 4))
    print("F1 Score :", round(f1_score(y_true_train, y_pred_train), 4))

    print("\nTesting Performance")
    print("Accuracy :", round(accuracy_score(y_true_test, y_pred_test), 4))
    print("Precision:", round(precision_score(y_true_test, y_pred_test), 4))
    print("Recall   :", round(recall_score(y_true_test, y_pred_test), 4))
    print("F1 Score :", round(f1_score(y_true_test, y_pred_test), 4))

    print("\nClassification Report (Test)")
    print(classification_report(y_true_test, y_pred_test))

# Evaluate logistic regression
evaluate_model(y_train, y_train_pred_log, y_test, y_test_pred_log, "Logistic Reg
```

```
===== Logistic Regression =====

Training Performance
Accuracy : 0.8343
Precision: 0.6813
Recall   : 0.4135
F1 Score : 0.5147

Testing Performance
Accuracy : 0.8371
Precision: 0.6864
Recall   : 0.43
F1 Score : 0.5287

Classification Report (Test)
              precision    recall  f1-score   support

           0       0.86      0.95      0.90      4207
           1       0.69      0.43      0.53      1135

    accuracy                           0.84      5342
   macro avg       0.77      0.69      0.72      5342
weighted avg       0.82      0.84      0.82      5342
```

Improved Model -Random Forest

Why use Rndom Forest model? It often perfoms better than logistic regression on complex datasets also it handles non-linear relationships well

In [17]:
```
# Create improved model pipeline
rf_model = Pipeline(steps=[
    ("preprocessor", preprocessor),
    ("classifier", RandomForestClassifier(
        n_estimators=200,
        max_depth=10,
        random_state=42
    ))
])

# Train the model
rf_model.fit(X_train, y_train)

# Predictions
y_train_pred_rf = rf_model.predict(X_train)
y_test_pred_rf = rf_model.predict(X_test)
```

Evaluate Random Forest

We evaluate the Random Forest so that we can compare its performance with Logistic Regression to determine whether the advanced model improves prediction quality.

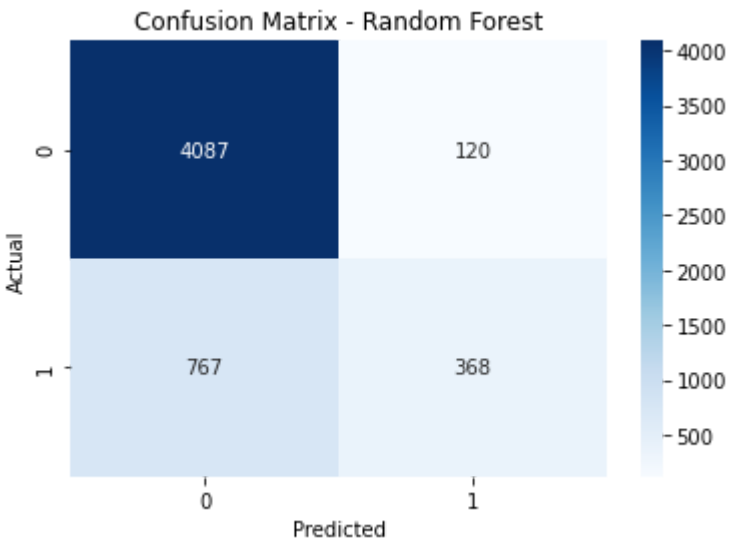
In [18]:
```
# Evaluate random forest
evaluate_model(y_train, y_train_pred_rf, y_test, y_test_pred_rf, "Random Forest"
```

```
===== Random Forest =====

Training Performance
Accuracy : 0.8586
Precision: 0.8649
Recall   : 0.3963
F1 Score : 0.5436

Testing Performance
Accuracy : 0.834
Precision: 0.7541
Recall   : 0.3242
F1 Score : 0.4535

Classification Report (Test)
              precision    recall  f1-score   support

           0       0.84      0.97      0.90      4207
           1       0.75      0.32      0.45      1135

    accuracy                           0.83      5342
   macro avg       0.80      0.65      0.68      5342
weighted avg       0.82      0.83      0.81      5342
```

Confusion Matrix

A confusion Matrix gives us a clearer picture of the types of errors the model makes, beyond accuracy alone

In [19]:
```
# Confusion matrix
cm = confusion_matrix(y_test, y_test_pred_rf)

plt.figure(figsize=(6,4))
sns.heatmap(cm, annot=True, fmt="d", cmap="Blues")
plt.title("Confusion Matrix - Random Forest")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()
```



Balance Check

To check whether the target classes are balanced.

In [20]:
```
# Check target distribution
print(df["h1n1_vaccine"].value_counts())
print(df["h1n1_vaccine"].value_counts(normalize=True))
```

```
0    21033
1     5674
Name: h1n1_vaccine, dtype: int64
0    0.787546
1    0.212454
Name: h1n1_vaccine, dtype: float64
```

Compare Models

This final table compares the two models on the testing data.

In [21]:
```
# Create a comparison table
results = pd.DataFrame({
    "Model": ["Logistic Regression", "Random Forest"],
    "Test Accuracy": [
        accuracy_score(y_test, y_test_pred_log),
        accuracy_score(y_test, y_test_pred_rf)
    ],
    "Test Precision": [
        precision_score(y_test, y_test_pred_log),
        precision_score(y_test, y_test_pred_rf)
    ],
    "Test Recall": [
        recall_score(y_test, y_test_pred_log),
        recall_score(y_test, y_test_pred_rf)
    ],
    "Test F1 Score": [
        f1_score(y_test, y_test_pred_log),
        f1_score(y_test, y_test_pred_rf)
    ]
})

print(results)
```

```
                Model  Test Accuracy  Test Precision  Test Recall  \
0  Logistic Regression       0.837140        0.686357     0.429956
1        Random Forest       0.833957        0.754098     0.324229

   Test F1 Score
0       0.528711
1       0.453481
```

Recommendation

1.Target groups with low vaccination likelihood. Use the model to identify individuals or population groups who are less likely to receive the H1N1 vaccine and prioritize them in awareness campaigns and outreach programs.

2.Strengthen doctor recommendations. Since doctor recommendation is likely a strong predictor of vaccine uptake, healthcare providers should be encouraged to actively recommend the H1N1 vaccine to eligible patients.

3.Use data-driven vaccination strategies. The predictive model can help public health planners allocate resources more effectively by identifying, high-risk groups, hesitant groups and regions or demographics with lower expected uptake

4.Focus on recall when evaluating interventions. Because missing people who are unlikely to vaccinate can weaken intervention efforts, stakeholders should pay close attention to recall when choosing the best predictive model.

Conclusion

This project developed a machine learning model to predict whether an individual received the *H1N1 vaccine* using survey data from the National 2009 H1N1 Flu Survey.The model helps public health stakeholders identify the factors that influence vaccine uptake and predict which groups are less likely to vaccinate. These insights can support targeted awareness campaigns, healthcare planning, and better vaccine distribution strategies.